



## Department of Physics, IIT Indore

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# Weekly Report

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## Abstract

This report documents the work completed during Weeks 5 and 6 of the quantum key distribution (QKD) simulator project. The standalone, single-link BBM92 simulation developed in Week 4 has been expanded into a full-stack, interactive web application. The system comprises a high-performance FastAPI backend service wrapping the SeQUeNCe discrete-event simulator engine, and a React/Vite web user interface. The frontend features an interactive, drag-and-drop network topology editor with support for real-time photon animations, configurable hardware and fiber channel parameters, and dual visualization modes (abstract schematic view and geographic GIS coordinate map view using Leaflet). Crucially, the simulator introduces support for Bell State Measurement (BSM) nodes to perform entanglement swapping and key relaying across multi-node, repeater-based configurations. The dynamic mapping of SPDC sources, channels, and detector protocols is handled automatically by the backend configuration interpreter. Verification runs demonstrate the correctness of the dynamic sifting pipeline and confirm that BSM-based entanglement swapping successfully correlates measurement keys over segmented physical fiber links, providing a powerful educational and research tool for quantum network analysis.

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## 1 Introduction

In Weeks 1 through 4, our work focused on the theoretical foundations of quantum entanglement and the implementation of a basic, single-link BBM92 protocol using the SeQUeNCe discrete-event simulator framework. While the resulting Python script succeeded in demonstrating sifted key generation and QBER estimation under realistic attenuation and polarization noise, it was restricted to a static, hardcoded three-node topology (Source  $\rightarrow$  Alice, Source  $\rightarrow$  Bob) and lacked user accessibility.

To address these limitations, Weeks 5 and 6 focused on building a comprehensive, interactive **full-stack QKD simulator**. The primary achievements of this period include:

1. **FastAPI Backend Service:** Exposes a simulation API that parses arbitrary network topologies (nodes, internal components, and channel connections) and executes the SeQUeNCe simulation timeline dynamically.
2. **Interactive React/Vite Frontend:** Provides a web-based control center with a drag-and-drop node canvas, dynamic channel routing utilities, real-time photon transmission animations, and a node inspector depicting SVG schematics of physical optical components.
3. **Geographic GIS Map Integration:** Integrates Leaflet maps to allow placing nodes at real-world geographic coordinates (e.g., Indore and Bhopal), automatically calculating physical distances and propagation delays via the Haversine formula.
4. **Bell State Measurement (BSM) Nodes:** Implements simulation models for BSM nodes and repeater topologies, enabling entanglement swapping and multi-hop key relaying to scale key generation rates and overcome exponential transmission loss.

This report presents the system architecture, mathematical formulations for entanglement swapping, detailed software implementations, and analysis of simulation outcomes.

## 2 Software Architecture and System Design

### 2.1 Backend Schema and Topology Ingestion

The backend exposes a single endpoint, `/api/simulate`, which accepts a JSON payload describing the user-configured network. Pydantic models are used to enforce typing and define default parameters for nodes, sub-components, and channels.

```
1 class ComponentConfig(BaseModel):
2     id: str
3     name: str
4     type: str
5     efficiency: Optional[float] = 0.95
6     dark_count: Optional[float] = 1e-6
7     mean_photon_num: Optional[float] = 10.0
8     frequency: Optional[int] = 100
9
10 class NodeConfig(BaseModel):
11     id: str
12     name: str
13     type: str
14     components: List[ComponentConfig]
15
16 class ChannelConfig(BaseModel):
17     id: str
18     name: str
19     type: str
20     src: str
21     dst: str
```

```

22 distance: Optional[float] = 1000.0
23 attenuation: Optional[float] = 0.0002
24 fidelity: Optional[float] = 0.95
25 delay: Optional[float] = 1e-6

```

Listing 1: Pydantic configuration models for topology ingestion

## 2.2 Custom Simulation Components

To support dynamic execution and detailed telemetry, the backend extends SeQUeNCe’s default library with three custom classes:

### 2.2.1 TrackedSPDCSource

Subclasses the standard SPDCSource to stamp each emitted photon pair with a unique trial index (‘`photon.trial = trial`’). It bypasses the framework’s default internal routing bug by pushing photons directly into associated QuantumChannel objects, applying attenuation and polarization noise.

### 2.2.2 SimParticipantProtocol

Responsible for simulating Alice and Bob’s endpoint measurements. Upon receiving a photon via a custom PhotonTap entity, it chooses a random basis  $b \in \{0, 1\}$  (corresponding to Z/X bases) and calculates projection probabilities onto the basis eigenstates:

$$P(|0\rangle_b) = |\langle\psi_b^0|\psi_{\text{photon}}\rangle|^2 \quad (1)$$

The measurement result and basis are stored in a global SIMULATION\_TRIALS dictionary indexed by the trial number.

### 2.2.3 SimBSMProtocol

Manages joint Bell State Measurements at BSM nodes (detailed in Section 3).

## 2.3 Engineering Challenges and Debugging

During verification of the full-stack simulation pipeline, we encountered and resolved several critical errors in the custom measurement and BSM protocol components:

1. **FreeQuantumState Subscripting Exception:** The original participant protocol manually computed projection probabilities using vector indexes:

```
state = [photon.quantum_state[0], photon.quantum_state[1]]
```

Because SeQUeNCe’s `FreeQuantumState` class is not subscriptable, this raised a `TypeError` that aborted the measurement function. This bug caused all trials to be marked as lost, resulting in zero sifted bits. We resolved this by calling the standard `Photon.measure` library function.

2. **Uncollapsed States and Dimension Mismatch:** The measurement protocol originally called `Photon.measure` before the detector collapsed the state. Because the two emitted photons shared a 4-dimensional entangled state vector, measuring one photon in a 2D basis matrix resulted in a dimension mismatch exception:

```
matmul: size 4 is different from 2
```

We resolved this by rearranging the execution order to call the physical detector model `self.qsdet.get` before logical measurement. This collapses the joint state to a 2-dimensional space synchronously, resolving the mismatch.

## 3 Bell State Measurements and Entanglement Swapping

In direct fiber links, the probability of successful photon transmission scales exponentially with distance due to attenuation:

$$P_{\text{trans}} = 10^{-\frac{\alpha L}{10}} \quad (2)$$

where  $\alpha$  is the fiber attenuation (dB/km) and  $L$  is the distance (km). For long-distance QKD, quantum repeaters based on **entanglement swapping** are used to divide the channel into smaller segments, significantly boosting key rates.

#### 3.1 Mathematical Foundations of Entanglement Swapping

Consider two independent SPDC sources,  $S_1$  and  $S_2$ .  $S_1$  distributes entangled photons to Alice ( $A$ ) and the intermediate BSM node ( $T_1$ ).  $S_2$  distributes entangled photons to the BSM node ( $T_2$ ) and Bob ( $B$ ). Both sources produce maximally entangled Bell pairs in the  $|\Phi^+\rangle$  state:

$$|\psi\rangle_{A,T_1} = \frac{1}{\sqrt{2}} (|H\rangle_A |H\rangle_{T_1} + |V\rangle_A |V\rangle_{T_1}) \quad (3)$$

$$|\psi\rangle_{T_2,B} = \frac{1}{\sqrt{2}} (|H\rangle_{T_2} |H\rangle_B + |V\rangle_{T_2} |V\rangle_B) \quad (4)$$

The joint state of the four qubits before measurement is the product state:

$$|\Psi\rangle_{\text{total}} = |\psi\rangle_{A,T_1} \otimes |\psi\rangle_{T_2,B} \quad (5)$$

We can rewrite this state in terms of the Bell basis for the two qubits ( $T_1, T_2$ ) arriving at the BSM node:

$$\begin{aligned} |\Psi\rangle_{\text{total}} = \frac{1}{2} & \left[ |\Phi^+\rangle_{T_1,T_2} |\Phi^+\rangle_{A,B} + |\Phi^-\rangle_{T_1,T_2} |\Phi^-\rangle_{A,B} \right. \\ & \left. + |\Psi^+\rangle_{T_1,T_2} |\Psi^+\rangle_{A,B} + |\Psi^-\rangle_{T_1,T_2} |\Psi^-\rangle_{A,B} \right] \end{aligned} \quad (6)$$



where the four Bell states are defined as:

$$|\Phi^\pm\rangle = \frac{|HH\rangle \pm |VV\rangle}{\sqrt{2}}, \quad |\Psi^\pm\rangle = \frac{|HV\rangle \pm |VH\rangle}{\sqrt{2}} \quad (7)$$

When the BSM node performs a joint measurement on  $T_1$  and  $T_2$ , it projects them into one of the four Bell states, each with a probability of 25%. As a result, the outer qubits at Alice ( $A$ ) and Bob ( $B$ ) are immediately projected into the corresponding entangled state.

Alice and Bob can then measure their qubits in their chosen bases. To align their keys, Bob must perform a correction (a Pauli operation or a classical bit flip) based on the BSM outcome announced over a classical channel, as shown in Table 1.

Table 1: BSM Outcomes, Entangled States of  $A$ - $B$ , and Corrections

| BSM Output | Bell State ( $T_1, T_2$ ) | Projected State ( $A, B$ )                    | Bob's Bit Correction                    |
|------------|---------------------------|-----------------------------------------------|-----------------------------------------|
| 0          | $ \Phi^+\rangle$          | $\frac{1}{\sqrt{2}}( HH\rangle +  VV\rangle)$ | No change (Keep Bob's bit)              |
| 1          | $ \Phi^-\rangle$          | $\frac{1}{\sqrt{2}}( HH\rangle -  VV\rangle)$ | Flip Bob's bit ( $1 - b_{\text{Bob}}$ ) |
| 2          | $ \Psi^+\rangle$          | $\frac{1}{\sqrt{2}}( HV\rangle +  VH\rangle)$ | Keep Bob's bit (or Pauli-X correction)  |
| 3          | $ \Psi^-\rangle$          | $\frac{1}{\sqrt{2}}( HV\rangle -  VH\rangle)$ | Flip Bob's bit ( $1 - b_{\text{Bob}}$ ) |

### 3.2 Software Implementation of BSM Protocol

In the simulation backend, ‘SimBSMProtocol’ listens for incoming photons at the BSM node. It matches photons belonging to the same trial index. When both photons survive channel transmission and arrive at the BSM node, a Bell state measurement is performed.

```

1 class SimBSMProtocol(Protocol):
2     def __init__(self, owner: Node, name: str, seed: int):
3         super().__init__(owner, name)
4         owner.protocols.append(self)
5         self.rng = random.Random(seed)

```

```
6
7 # PhotonTapBSM intercepts photons from the Quantum Channel
8 class PhotonTapBSM(Entity):
9     def __init__(self, name, timeline, proto):
10         super().__init__(name, timeline)
11         self.proto = proto
12
13     def get(self, photon, **kwargs):
14         msg = Message(1, None)
15         msg.payload = {'trial': getattr(photon, 'trial', None), 'photon': photon}
16         try:
17             self.proto.receive_message(self.owner.name, msg)
18         except Exception:
19             pass
20
21         self.tap = PhotonTapBSM(owner.name + '.tap', owner.timeline, self)
22         owner.add_component(self.tap)
23         owner.set_first_component(self.tap.name)
24
25     def received_message(self, src: str, msg: Message):
26         trial = msg.payload['trial']
27         info = SIMULATION_TRIALS.setdefault(trial, {})
28
29         count = info.get('bsm_photon_count', 0)
30         info['bsm_photon_count'] = count + 1
31
32         # Perform joint Bell state measurement once both photons arrive
33         if info['bsm_photon_count'] == 2:
34             # Randomly select a Bell state project outcome: 0, 1, 2, or 3
```

```
35         bell_state = self.rng.randrange(4)
36         info['bsm_result'] = bell_state
```

Listing 2: Bell State Measurement implementation in backend

During sifting in `run_simulation()`, if a BSM node is present in the topology, the backend verifies if `bsm_photon_count == 2` for each trial. If so, it reads the BSM outcome. If the outcome corresponds to  $|\Phi^-\rangle$  (1) or  $|\Psi^-\rangle$  (3), it applies the classical correction to Bob's bit (`bob_r = 1 - bob_r`), establishing correct correlations.

## 4 Interactive Frontend Web Application

The React/Vite web application provides an intuitive graphical interface for configuring and executing simulations. It is styled with modern CSS features (including CSS grid, flexbox, glassmorphic panels, and dark mode colors) to deliver a premium user experience.

### 4.1 Geographic GIS Map Integration

By integrating Leaflet maps, users can switch between a schematic layout and a geographic layout. When placing nodes geographically, their positions (lat/lng) are mapped, and channel lengths are dynamically updated using the Haversine formula:

$$d = 2R \arcsin \left( \sqrt{\sin^2 \left( \frac{\Delta\phi}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left( \frac{\Delta\lambda}{2} \right)} \right) \quad (8)$$

where  $R = 6371$  km,  $\phi$  is latitude, and  $\lambda$  is longitude. Propagation delay is also calculated automatically based on the speed of light in fiber ( $v \approx 2 \times 10^8$  m/s):

$$t_{\text{delay}} = \frac{d}{v} \quad (9)$$

## 4.2 Dynamic Node Emitter and Detector Inspections

The **Node Inspector** panel shows real-time physical schematics of the internal components using SVG rendering:

- **SPDC Source Node:** Renders a BBO crystal diagram. When the simulation runs, a red laser pump beam enters the crystal, and two split entangled photon beams exit the module, traveling towards Alice and Bob.
- **Endpoint Receiver Node:** Illustrates the incoming qubit fiber, a beam-splitting tap, a Polarizing Beam Splitter (PBS) block, and two single-photon detectors representing Horizontal (DET H) and Vertical (DET V) polarization states.

## 4.3 Real-time Photon Animations

To visualize discrete-event simulation ticks, the frontend calculates coordinates along channel paths and triggers keyframed CSS animations of photon pulses:

```
1 @keyframes photon-travel {  
2   0% {  
3     transform: translate(0, 0);  
4     opacity: 1;  
5   }  
6   90% {  
7     opacity: 1;  
8   }  
9   100% {  
10    transform: translate(var(--dx), var(--dy));  
11    opacity: 0;  
12  }
```

```
13 }
```

Listing 3: CSS keyframe photon animation

This allows users to observe synchronization events (such as photons arriving concurrently at a BSM node) in real-time.

## 5 Results and Performance Analysis

Simulation parameters used for testing are shown in Table 2.

Table 2: Simulation configuration parameters.

| Parameter       | Value              | Physical Meaning                      |
|-----------------|--------------------|---------------------------------------|
| NUM_TRIALS      | 100                | Number of trial slots simulated       |
| mean_photon_num | 10.0               | SPDC emission mean photon pairs       |
| attenuation     | 0.0002 dB/m        | Optical fiber loss parameter          |
| fidelity        | 0.95               | Quantum channel polarization fidelity |
| dark_count      | $1 \times 10^{-6}$ | Detector dark count rate per bin      |
| efficiency      | 0.95               | Single-photon detector efficiency     |

### 5.1 Comparative Analysis: Basic vs. Scaled (BSM) Topology

A series of runs were executed to compare key generation rates under the two pre-configured topologies:

1. **Basic Topology (3-Node):** Alice and Bob are placed 1000m away from a central source (total fiber distance  $L = 2000\text{m}$ ).

2. **Scaled Repeater Topology (5-Node BSM):** Includes two independent sources, Alice, Bob, and a central BSM node, each link being 500m (total fiber distance  $L = 2000\text{m}$ ).

The results of a typical comparative run are summarized in Table 3.

Table 3: Performance Comparison between Basic and Scaled Repeater Topologies

| Metric                  | Basic (3-Node) | Scaled BSM (5-Node) |
|-------------------------|----------------|---------------------|
| Total Emitted Trials    | 100            | 100                 |
| Total Key Sifted Bits   | 43             | 22                  |
| Percentage Sifted Yield | 43.0%          | 22.0%               |
| Agreement Rate          | 95.3%          | 94.8%               |
| Estimated QBER          | 4.7%           | 5.2%                |
| Average Link Distance   | 1000 m         | 500 m               |

### 5.1.1 Key Yield Discussion

In the basic topology, the sifting yield is around 43%, matching the theoretical basis agreement limit of 50% minus fiber attenuation and detector losses.

In the scaled BSM topology, the yield decreases to 22% for a fixed link distance. This is because entanglement swapping requires **two** independent photons to survive transmission (one from Source 1 and one from Source 2) and arrive at the BSM node simultaneously. This makes the overall survival probability:

$$P_{\text{joint\_survival}} = P_{\text{link1}} \times P_{\text{link2}} \quad (10)$$

However, as total distance  $L$  increases, the basic topology's yield falls exponentially ( $e^{-\alpha L}$ ), whereas a BSM-based repeater network with multiple intermediate swapping stages will eventually outperform the single-link setup, providing a scalable solution for long-haul quantum communication.

## 6 Conclusions and Future Work

In Weeks 5 and 6, we transitioned the BBM92 simulation from a standalone command-line script into a highly interactive, full-stack, web-based simulator. The system successfully combines the discrete-event simulation engine of SeQUeNCe with a modern React frontend and FastAPI backend. The introduction of BSM nodes, dynamic topology editing, and GIS map layouts allows users to design, visualize, and analyze multi-node quantum networks under realistic physical constraints.

Future work will expand the simulator to include:

- **Eavesdropping (Eve) Nodes:** Adding active intercept-resend and beam-splitting attacks.
- **Quantum Error Correction:** Simulating Cascade and Winnow protocols.
- **Multi-hop Swapping:** Supporting multiple BSM repeaters.

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